29 Ans: D

When the number of undecayed nuclei in the two sources are equal, $N_1 = N_2$

$$N = \left(\frac{1}{2}\right)^{\frac{t}{t_{1.2}}} N_0$$

$$\left(\frac{1}{2}\right)^{\frac{t}{2}} (64) = \left(\frac{1}{2}\right)^{\frac{t}{3}} (8)$$

$$\left(\frac{1}{2}\right)^{\frac{t}{2}-\frac{t}{3}} = 2^{3-6}$$

$$\frac{t}{6} = 3$$

$$t = 18 \text{ days}$$

30 Ans: D